



OBJECT ORIENTED SOFTWARE ENGINEERING

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2. Class Based Model

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- Class Modeling in software engineering is the process of simplifying the diagram or data model of a software system by applying certain formal techniques.
- It involves expressing data and information through **text and symbols**.
- The data/class model provides the **blueprint for building** a new system or reengineering legacy applications.

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UML Model - building blocks



The building blocks of UML Models can be defined as:

- **Entities**
- **Relationships**
- **Diagrams**

- **Entities are the elements of models.**
- All UML entities can be divided into 3 major entities categories. They are:
 1. **Structural entities – nouns**, UML models, such as class, interface. These are the static parts of a model, representing elements that are either conceptual or physical.
 - Class and Objects
 - Interface
 - Collaboration
 - Use case

2. Behavioral entities – verbs, UML models, such as interaction, activity, State machines; grouping the essence of the package that is used for grouping related elements of the model in forming a single whole modules;

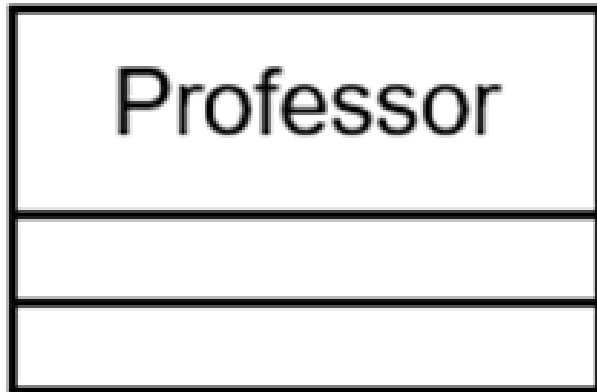
3. Summary entity - note, which can be added to the model to record specific information, very much like the **sticker**

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Representing Class in UML



A class is represented using a compartmented rectangle



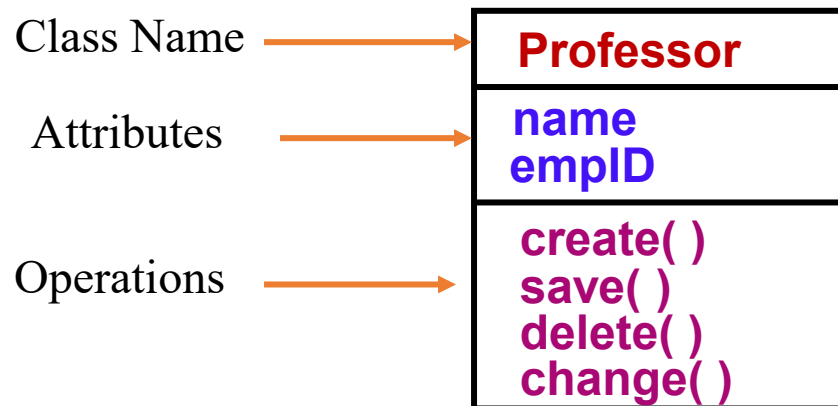
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Representing Class in UML



A class is comprised of three sections

- The **first section** contains the **class name**
- The **second section** shows the **structure (attributes)**
- The **third section** shows the **behavior (operations)**

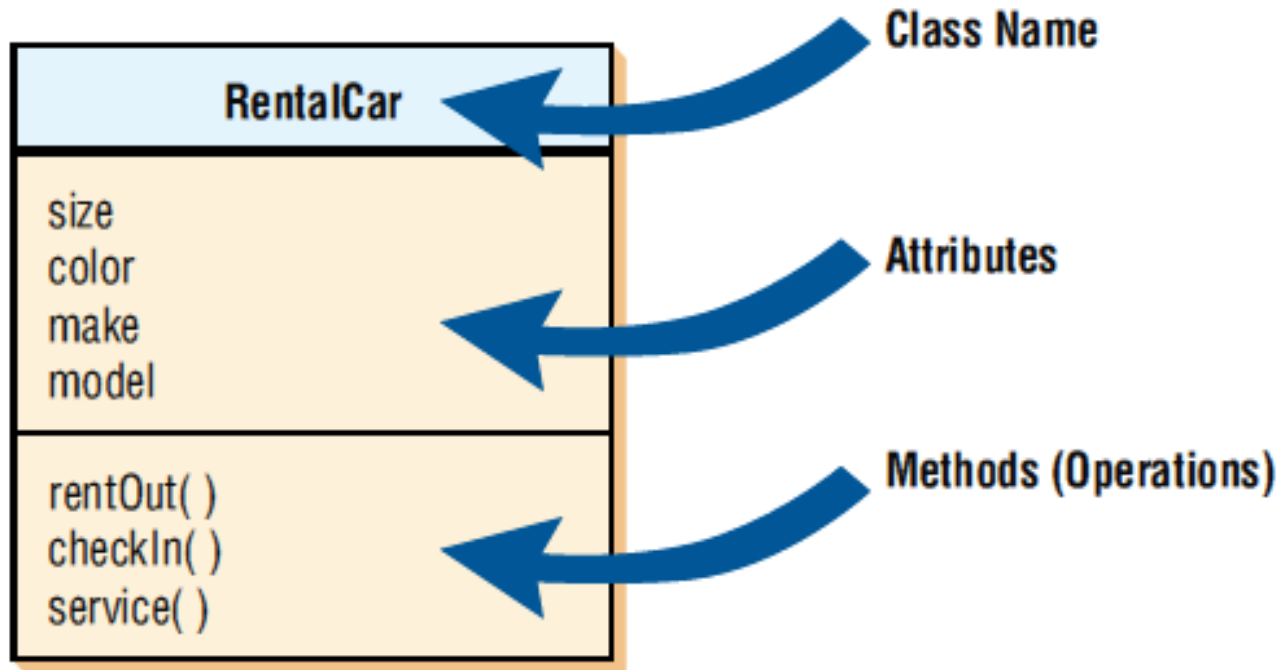


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Representing Class in UML



Consider an example: for **RentalCar** Class, the class diagram representation is as shown below:

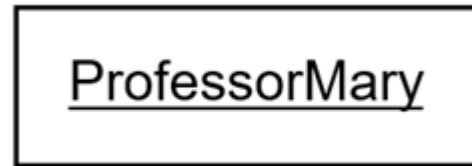


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Representing Objects in UML

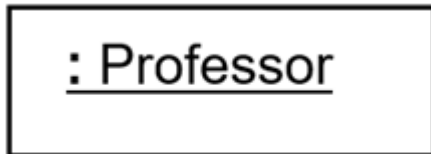


An object is represented as rectangles with **underlined** names

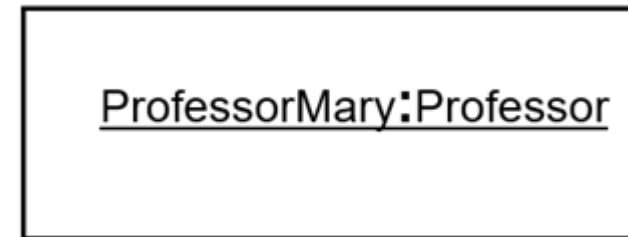


Object Name Only

The other forms of object representations are:



Class Name Only



Object Name with its Class Name

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UML Relationships

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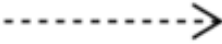
The most frequently used relationships in UML are:

1. Dependency
2. Association
3. Generalization
4. Realization

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Relationships in UML

The following are the list of relationships in UML :

Type of relationship	UML syntax Source – Target	Semantics
Dependency		Source element depends on the target element and change last may affect the first.
Association		Description of the set of relationships between objects.
Aggregation		The target element is a part of the source element.
Composition		Strict (more limited) form of aggregation/

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Relationships in UML

Type of relationship	UML syntax Source – Target	Semantics
Inclusion		Source element contains the target element.
Generalization		The source element is a specialization of a more generalized the target element, and can replace him.
Realization		The original item is guaranteed to perform the contract, a certain target element.

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UML Diagrams

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Class Diagram:

- Class Diagram gives the **static view** of an application.
- A class diagram describes the **types of objects** in the system and the different types of **relationships that exist** among them.
- A class can refer to another class. A class can have its objects or may inherit from other classes.

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Categories of UML Diagram - Class Diagram...

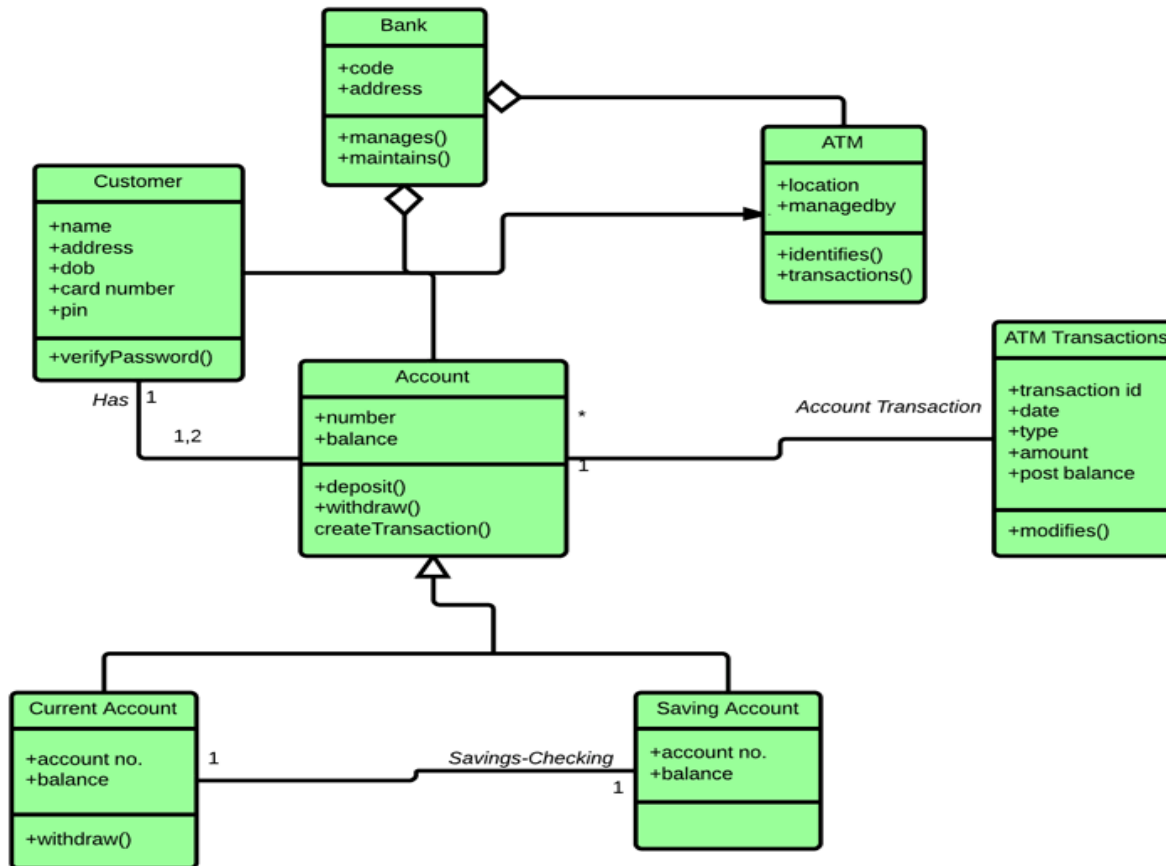


- UML Class Diagram gives an **overview** of a software system by displaying **classes, attributes, operations, and their relationships**.
- This Diagram includes the **class name, attributes, and operation** in **separate designated compartments**.
- Class Diagram helps construct the code for the software application development.
Essential elements of UML class diagram are:
 - **Class Name**
 - **Attributes**
 - **Operations**

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Categories of UML Diagram - Class Diagram...

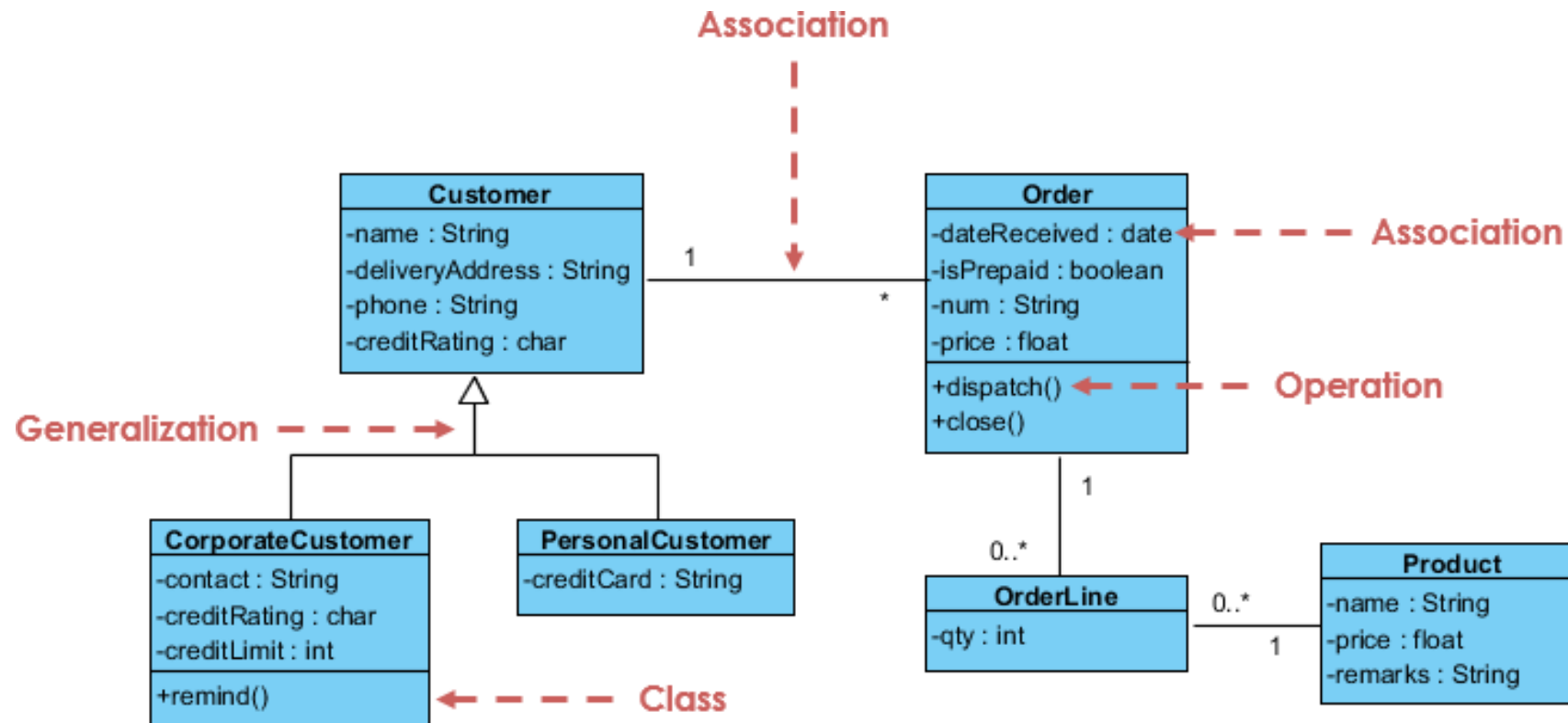
- UML Class Diagram example 1



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Categories of UML Diagram - Class Diagram...

- UML Class Diagram example 2





THANK YOU

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